Name_____ Date_____ Period_____

LESSON 8-5 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Graphing Inequalities

Maryland Standard 6.EE.B.8 · Learning target – I can graph an inequality on a number line using an open or a closed dot.

REMEMBER

- Use an **open dot** for < or >. Use a **closed dot** for ≤ or ≥.
- Draw the arrow toward the solutions. The dot depends on the **symbol**, not on the sign of the number.
- Test a number under your arrow. It should make the inequality true.

WORKED EXAMPLE

Graph $y > -2$.



Step 1 > does not include −2, so draw an open dot at −2.

Step 2 Solutions are greater, so the arrow points right.

Answer **Open dot at −2, arrow right**

WORD BANK

open dot – the number is NOT a solution (< or >)

closed dot – the number IS a solution (≤ or ≥)

at least – that number or more (≥)

arrow – points to all the numbers that work

PRACTICE

use the number lines · show your work

1. For each graph, tell the dot (open or closed) and the arrow direction (left or right).

Step 1 Look at the symbol: < or > means open; ≤ or ≥ means closed.

Step 2 Greater (> or ≥) points right. Less (< or ≤) points left.

a. $x \geq 9$ _____

b. $n < -4$ _____

c. $p \leq 0$ _____

d. $k > 2.5$ _____

e. $w \geq -6$ _____

f. $r < 7$ _____

g. $b > -1.5$ _____

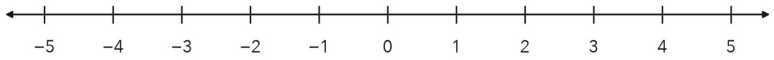
h. $c \leq 12$ _____

2. Graph $a \leq -1$ on the number line. Then name one solution.

Step 1 Find −1 on the number line.

Step 2 ≤ includes −1, so choose the dot.

Step 3 Shade toward the numbers that are less.

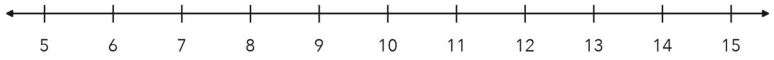


One solution_____

3. **Hiking Trail.** Hikers must be **at least** 10 years old. Write an inequality for *a*, a hiker's age. Then graph it.

Step 1 "At least 10" means 10 or more.

Step 2 Draw the dot at 10, then the arrow.



Inequality_____

4. **Read the Graph.** Write the inequality shown, using *m*. Then describe a real situation it could show.

Frame *The graph shows m ___ 6. A real situation is ___.*



_____

_____

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Ask someone at home for a rule with a limit (bedtime, screen time, a speed limit). Write it as an inequality and graph it.

Name_____ Date_____ Period_____

LESSON 8-5 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Testing Solutions to Inequalities

Maryland Standard 6.EE.B.8 · Learning target – I can test values against an inequality and graph the solution set.

REMEMBER

- **Substitute** the value. A **true** statement means it is a solution.
- An inequality usually has **many** solutions.
- The **boundary** is a solution for ≤ and ≥ only.

WORKED EXAMPLE

Which of 6, 8, and 11 are solutions of $x \geq 8$?

Step 1 Test 6: $6 \geq 8$ is false. Test 8: $8 \geq 8$ is true.

Step 2 Test 11: $11 \geq 8$ is true.

Answer **8 and 11 are solutions; 6 is not.**

WORD BANK

substitute – put the number in place of the letter

solution – a number that makes it true

boundary – the number in the inequality

true – the comparison is correct

PRACTICE

show your work · label units

1. Substitute each value. Write **yes** if it is a solution or **no** if it is not.

Step 1 Put the value in place of x.

Step 2 Read the comparison. True means yes; false means no.

Inequality	$x = 4$	$x = 7$	$x = 10$
$x > 7$			
$x \leq 7$			
$x \geq 4$			

2. Is the value a solution? Write **yes** or **no**.

a. $x < 9$; $x = 9$ ___ **b.** $m \geq -1$; $m = 0$ ___ **c.** $t \leq 2.5$; $t = 3$ ___

3. **Bowling.** A bowling team wins a trophy if it scores **more than** 450 points. Write an inequality for the score s. Test 450, 445, and 472.

Step 1 "More than" means >.

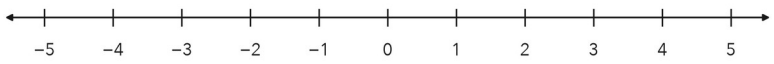
Step 2 Put each score in place of s. Is it true?

Inequality and winning scores_____

4. Test −3, 0, and 2 in $y \leq -1$. Then graph $y \leq -1$ on the number line.

Step 1 Put −3, 0, and 2 in place of y, one at a time.

Step 2 ≤ includes −1, so choose the dot. Then draw the arrow.



Solutions from the list_____

5. Find the Error. Bianca says $n < 12$ has 12 solutions: 11, 10, 9, and so on down to 0. Explain what she is missing.

Frame *Bianca is missing ___. For example, ___ is also a solution.*

_____

_____

TEST PRACTICE

choose the best answer

6. Which value is **not** a solution of $k \geq -2$?

A −3 **B** −2 **C** 0 **D** 5

7. Which value is a solution of $n > 15$?

A 12 **B** 14.9 **C** 15 **D** 15.5

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___